

Estimating receptive vocabulary from incidental reading behavior

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Abstract

Language learning platforms typically estimate vocabulary through placement tests, which quickly go stale, or manual word marking, which is tedious and error-prone. We present a method that continuously and non-intrusively estimates receptive vocabulary from reading behavior. When learners can click any word for a translation, we observe both the words they translate and those they read without translating: a word met several times without a lookup is likely already known.

Using longitudinal data from Zeeguu, an open-source reading platform, we analyzed the reading of 322 learners of French, Dutch, English, German, and Danish over months to years. Of the words learners look up, 74% are looked up in the first article in which they appear. A word left untranslated in three or more articles is not looked up again in 96% of cases, strong behavioral evidence that it is known. We turn this into a per-word probability of knowledge; aggregated over a learner's reading, it yields a continuously updated vocabulary estimate that supports practice prioritization and motivational feedback. To our knowledge, no prior work treats the absence of a dictionary lookup as evidence of word knowledge.

Keywords: vocabulary estimation; incidental vocabulary learning; reading behavior; implicit assessment.

1. Introduction

Vocabulary acquisition is fundamental to second language learning, yet most language learning systems lack reliable estimates of what words a learner already knows. The stake is practical: lexical coverage of roughly 95 to 98% of the running text in a passage is tied to adequate and to unassisted comprehension respectively (Hu & Nation, 2000; Nation, 2006; Laufer & Ravenhorst-Kalovski, 2010), so an estimate of a learner's vocabulary is, indirectly, an estimate of which texts they can read unaided. Current approaches to obtaining that estimate have significant limitations. **Placement tests** (e.g., Duolingo) give a one-time snapshot. Learners keep acquiring vocabulary outside the app (through reading, conversation, media), so the test goes stale and they are tested on words they learned months previously. **Manual word marking** (e.g., LingQ, Readlang) asks learners to flag what they know. It is tedious and error-prone: LingQ marks a page's remaining words as "known" when the learner moves on, so users report accidentally inflating their counts by hundreds.¹ Demanding an action for every word also interrupts reading.

No current system, to our knowledge, infers vocabulary knowledge from what learners *don't* do. This is exactly what we propose: if a learner encounters a word across multiple articles without ever requesting a translation, they

¹ LingQ Language Forums, "Someone please fix" (2022), <https://forum.lingq.com/t/someone-please-fix-paging-moves-to-known/35718>

most likely already know it. Such an implicit signal enables continuous vocabulary profiling without interrupting the reading experience.

2. The Zeeguu Platform

Zeeguu (zeeguu.org) is an open-source platform (github.com/zeeguu) for learning a language by reading (Lungu et al., 2018). Rather than following a fixed syllabus, it recommends authentic web-sourced articles (news, blog posts, etc.) in the learner's target language, selected to fit their interests and their proficiency. Titles and summaries are rewritten for every level, and an article that is too difficult can be simplified on demand.

As the learner reads, any unfamiliar word can be translated in place with a single click, or a multi-word phrase by clicking each of its words in turn to extend the selection; every lookup is recorded. The words a learner has translated feed vocabulary exercises, ordered by each word's frequency in the target language, as well as generated listening lessons. Zeeguu runs as a web app, as iOS and Android apps, and as a browser extension, and currently has several hundred monthly active users reading across 11 languages.

3. Related Work

Our approach draws on three research streams: vocabulary size estimation, incidental vocabulary acquisition through reading, and, closest to our own contribution, work that infers word knowledge from implicit behavioral signals.

Vocabulary Size Estimation. Estimating vocabulary size has long been central to language assessment research. Traditional approaches include explicit tests such as the Vocabulary Size Test (Nation & Beglar, 2007) and Yes/No format tests pioneered by Meara and colleagues (Meara & Buxton, 1987; Meara & Jones, 1990). In Yes/No tests, learners indicate whether they recognize words as real vocabulary items, with pseudowords included to detect guessing. While efficient, these tests are one-off snapshots that interrupt the learning flow and are not designed for continuous, unobtrusive progress tracking. Like them, our method measures receptive knowledge; unlike them, it does so passively and continuously, as a by-product of reading. Closer to a maintained estimate is Lee (2022), whose text-recommendation system holds an explicit per-word model of what a learner knows and ranks texts by their share of unknown words. It is editable by design: the learner inspects and corrects the model, buying transparency at the cost of the upkeep our method does not require.

Vocabulary Acquisition and Dictionary Lookups. How much exposure it takes to *acquire* a word varies widely, from six to more than twenty encounters (Webb, 2007), but that is a different question from ours: we ask whether a word is *already* known. Closer to our interests is research on dictionary lookup behavior during reading, which shows that looking a word up is driven by more than not knowing it: dictionary design shapes what a learner takes from a lookup (Laufer & Hill, 2000), and topic interest shapes how they look words up (Cancino, 2023). A lookup is therefore a noisy sign of ignorance, and its absence, on its own, only a tentative indication of knowledge.

Implicit Signals of Word Knowledge. Closest to our own approach is work that infers word knowledge from reading behavior rather than from responses to test items. Eye-tracking research has long established that fixation duration is sensitive to word familiarity and frequency, and that frequent, predictable words are more often skipped altogether (Rayner, 1998). In L2 vocabulary research specifically, Godfroid, Boers, and Housen (2013) showed that learners spend longer processing unknown words than matched known controls, and that the longer they look, the more likely they are to recognize the word afterwards. Godfroid et al. (2018) extended this to natural reading and found that reading times fall along an S-shaped curve as a word is met again and again, with the number of exposures the strongest predictor of whether it is learned.

These studies show that reading behavior reveals word knowledge; where we differ is instrumentation and scale.

Eye-tracking needs laboratory hardware and yields minutes of reading per participant, whereas a translation click is logged by the platform itself, for every learner, potentially across years of ordinary reading.

A second stream, knowledge tracing, models a learner’s mastery over time from interaction logs (Corbett & Anderson, 1995; Piech et al., 2015). These models are well established, but they trace mastery from responses to items the system has posed: exercises, quizzes, and flashcard reviews. Our evidence is of a different kind: it comes from moments when the learner was asked nothing and did nothing. Treating that absence as positive evidence of knowledge is, to our knowledge, new.

4. Method

Our data is the interaction record the platform already keeps, with no special instrumentation added for this study. The input comes from two kinds of logged event:

1. **Reading session.** A learner opens an article and reads it. The platform records the article, when the session started, how long the article was open, and the word lookups made during it. Returning to the article later starts a new session.
2. **Word lookup.** A learner clicks an unfamiliar word for a translation. The platform records it together with its translation, context sentence, article, and time. We count only lookups made while reading, not in exercises or article previews. A word looked up more than once within a single reading session counts once. We do not count multi-word lookups in this paper.

Because every word in a read article counts as evidence, positive or negative, we define a read article carefully.

4.1 Read Articles

Not every opened article is actually read: learners skim, read only part of it, or open it, glance through, and decide it is not for them. We count an article as **read** when the learner spent enough time to have genuinely read it: *at a pace no faster than 300 words per minute, and for at least 30 seconds*, i.e. at least $\max(30s, n \times 200ms)$ for an article of n words. The speed ceiling rules out skimming, and is deliberately generous: adult first-language silent reading of English non-fiction averages 238 wpm, with most readers between 175 and 300 (Brysaert, 2019), and second-language reading is slower still. The 30-second floor guards against accidental opens of short articles rather than making a reading-rate claim; where it binds tighter than the ceiling, it costs 0.50% of otherwise qualifying reading (155 of 30,716).

Time is summed across the learner’s sessions on that article. For most of our data it is wall-clock time, so an article left open in a background tab keeps accumulating; the platform has since begun pausing the timer on inactivity.

4.2 Computing $P(\text{known})$ for Each Word

By *knowing* a word we mean receptive form–meaning knowledge in Nation’s (2013) sense: recognizing the written form and retrieving the meaning it has in the text being read. Other aspects of word knowledge, and other senses of the same word, are beyond what reading behavior can show (§7).

We estimate knowledge of each word incrementally, replaying the learner’s history chronologically. Each word (in practice each stem, §4.4) has a single score S that every event adjusts:

- a **clean encounter** (the word appears in a read article and is not looked up there; an article counts once, however often the word occurs in it) is evidence that the word is known;
- a **lookup** is evidence that it is not.

We hold S in *log-odds*, the scale on which successive pieces of evidence simply add up, as in logistic learner models (Pelánek, 2017), and which we convert back into a probability with the S-shaped function $\sigma(x) = 1/(1 + e^{-x})$. Because each event only nudges one number, the estimate can be maintained live as a learner reads, without ever recomputing over their past. Older evidence is discounted at each step, so recent behavior weighs more:

$$S \leftarrow \lambda S + w_{\text{clean}}, \quad S \leftarrow \lambda S - w_{\text{lookup}}, \quad P(\text{known}) = \sigma(S).$$

Before any evidence $S = 0$, so $P(\text{known}) = 0.5$, the prior. Because evidence accumulates in log-odds, the probability moves in shrinking steps rather than by a fixed amount (one clean encounter gives $P(\text{known}) \approx 0.62$, three ≈ 0.79 , five ≈ 0.89 , etc). We set a lookup ($w_{\text{lookup}} = 1.0$) to weigh more than a clean encounter ($w_{\text{clean}} = 0.5$) because a lookup is an explicit signal of not knowing, whereas a clean encounter may also reflect skimming or guessing from context. With $\lambda = 0.9$, an early lookup fades as the learner goes on to read the word cleanly, and a once-fluent word looked up again after a long gap drops back, so estimates can fall as well as rise.

Two of these choices are deliberate simplifications: (1) the decay is applied per event rather than per unit of elapsed time, so two encounters weigh the same whether minutes or months apart; and (2) w_{clean} , w_{lookup} , and λ are set by hand rather than fit to data. Both are revisited in §9.

4.3 Vocabulary Bands

We analyze coverage across nested frequency bands, ranked with the *wordstats* library, whose frequency lists are built from film subtitles in each language (§7):²

- **Top 100:** function words, pronouns, basic verbs.
- **Top 500:** common nouns, adjectives, verbs.
- **Top 1000:** essential vocabulary for basic communication.
- **Top 5000:** intermediate to advanced vocabulary.

4.4 Aggregation into Coverage, Vocabulary Size, and CEFR

Word-level $P(\text{known})$ estimates are aggregated per learner at monthly snapshots. Word forms are stemmed with Snowball (Porter, 2001) before scoring, so each score belongs to a stem, a near-lemma rather than a raw inflected form, and the counts stay comparable to the lemma-based CEFR bands. We count a stem as *known* when $P(\text{known}) \geq 0.75$, which takes three clean encounters with no offsetting lookups, a deliberately conservative threshold (§6.1).

Both *band coverage* and *estimated vocabulary size* are extrapolated from what the learner has been observed to meet, the same way a vocabulary size test extrapolates from the items it samples. Within each block of 100 consecutive frequency ranks we take the share of the stems the learner has *encountered* that are known, and apply that rate to the whole block; blocks in which fewer than five stems have been met are credited nothing rather than guessed at. Band coverage is the resulting share of a band’s words that a learner knows (not a share of running text). Estimated vocabulary size is the same measure applied to the 5,000 most frequent stems, reported as a count (at most 5,000) rather than a share.

We map the result to an indicative CEFR level using the published receptive vocabulary-size bands of Milton and Alexiou (2009). Those are calibrated on tests that sample the whole frequency range, whereas our evidence reaches only as far as a learner has read, so the level we report is a floor rather than an estimate.

²github.com/zeeguu/python-wordstats, built on frequency lists derived from OpenSubtitles. Subtitle frequencies are a standard choice: they predict word recognition better than frequencies from written corpora (Brysbaert & New, 2009), and they come from one comparable source for every language we study.

4.5 Learner Sample

We analyzed longitudinal data from 322 learners (324 learner-language profiles, as two read in two target languages) who used Zeeguu for periods ranging from several months to over two years. All had read at least 20 articles, in the sense of §4.1. Developer and test accounts are excluded, as are target languages read by ten or fewer learners, too few to form a sample. Profiles per target language are given in Table 1.

French	Danish	Dutch	German	English	Total
264	18	17	14	11	324

Table 1. Learner-language profiles per target language.

5. Results

Our method depends on the assumption that learners look up the words they do not know, and do so early, usually the first time they meet them. If so, a word that has appeared in several articles without a lookup is probably known, and the earlier lookups come, the fewer articles we need to see before treating a word as known.

5.1 When a Word Is Translated, How Early?

Across all 324 profiles and their 122,303 translated words (each word form counted once per learner), we measured, for each one, in which of the learner’s read articles containing it the first translation came (Table 2).

Language	Words	by 1st article	by 2nd	by 3rd	by 4th	by 5th
English	1,017	53.1%	94.7%	98.1%	99.0%	99.2%
Dutch	12,649	65.7%	94.8%	97.7%	98.6%	99.0%
German	4,298	88.6%	95.8%	97.7%	98.7%	98.9%
Danish	5,993	85.2%	94.6%	97.2%	98.2%	98.6%
French	98,346	73.5%	91.3%	95.3%	97.0%	97.8%
All	122,303	73.6%	92.0%	95.7%	97.3%	98.1%

Table 2. Share of translated words first looked up by the 1st to 5th article containing them (cumulative).

Languages vary in how readily learners look a word up on first encounter (from 53% for English learners to 89% for German, 74% overall, though the smaller-language figures rest on a handful of learners each). However, by the second article every language reaches over 91% for all, and by the fifth almost all translated words have been looked up (overall 98%).

These figures cover only words learners did look up, so they test the method’s assumption only in part: unknown words never looked up (inferred from context, guessed, or read past) leave no trace and are counted as known (§7).

5.2 Vocabulary Coverage Patterns

Table 3 shows four active learners at their most recent snapshot.

Learner	Language	Articles	Top 100	Top 500	Top 1000	Words known	CEFR
A	Danish	20	92%	59%	37%	503	A1
B	English	79	97%	88%	78%	2,466	A2
C	French	88	94%	89%	79%	2,318	A2
D	Dutch	2,854	100%	99%	98%	4,374	C1

Table 3. Band coverage (share of each band’s stems known), estimated vocabulary size and indicative CEFR level for four different learners.

Across these learners, coverage rises with reading volume: the Top 100 band fills first, while the Top 1000 band lags and keeps climbing. Learners C and D are plotted in Figure 1.

5.3 Monthly Estimates

The monthly snapshots (§4.4) become informative with modest reading: Learner C (Figure 1), reading about seven articles a month, shows band coverage climbing month to month; heavier reading in mid-2020 mainly deepens the less frequent bands.

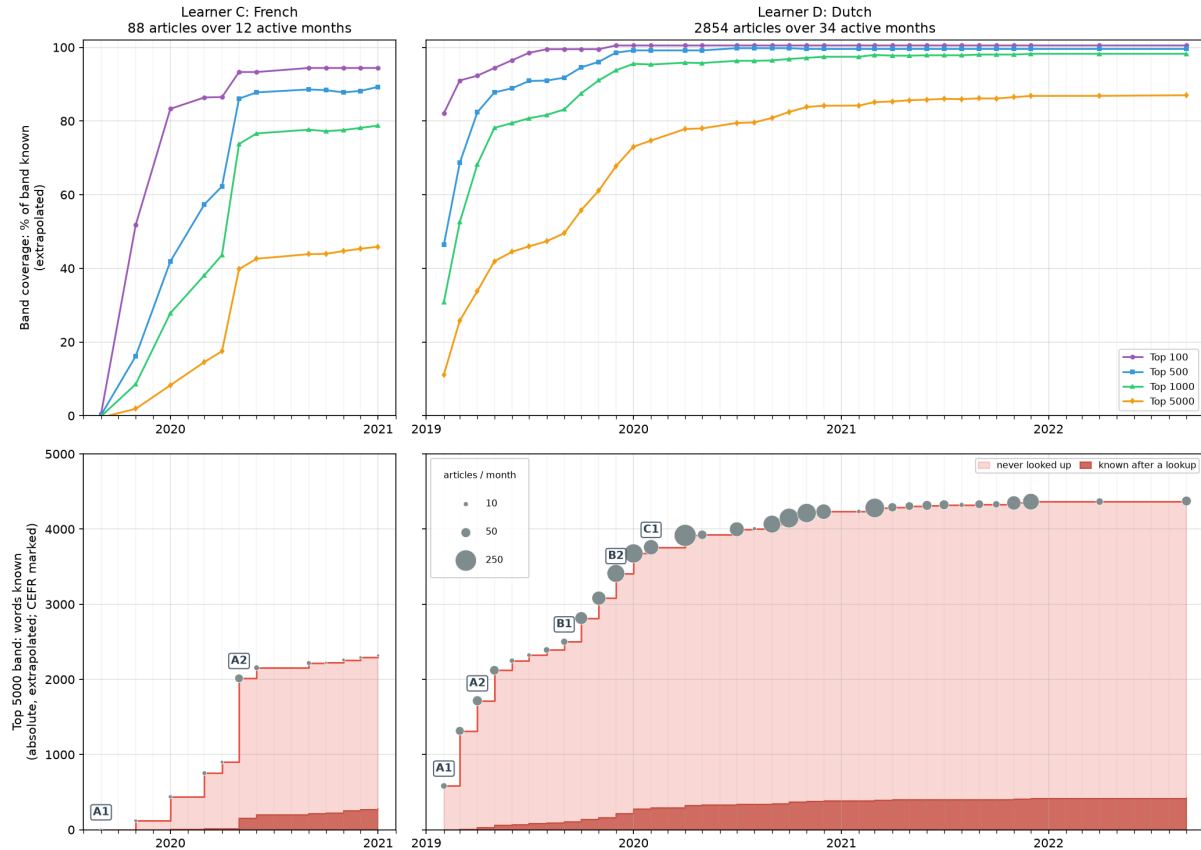


Figure 1. Monthly vocabulary estimates for two learners, computed by the $P(\text{known})$ model. Left: Learner C (French) read 88 articles over 12 active months, A1 to A2. Right: Learner D (Dutch) read 2,854 articles over 34 active months, A1 to C1. The lower panels show the Top 5000 band as a word count, the range the CEFR mapping uses (§4.4). Both columns share one time scale. In the lower panels, dot area is proportional to articles read that month: the estimate changes with reading and stays flat in months without it. The darker band at the bottom marks words known after an earlier lookup (plausible acquisition); the lighter area above it, words never looked up (prior knowledge being discovered, or words learned without help from the application).

Learner C also shows that CEFR level is far coarser than the bands: moving from A1 to A2 takes more than half a year. Across the dataset only 60 of 324 profiles cross a level, and just one crosses more than two; band coverage, not CEFR level, tracks a modest reader’s progress.

Learner D is an outlier: 2,854 articles, roughly 100 times the median of 27 and more than ten times the next most active reader, so the A1→C1 trajectory is exceptional. Volume determines how much vocabulary the estimate can see, not how reliable it is. But the method can only judge words a learner has met, so a light reader’s estimate is best read as a conservative lower bound.

A rising estimate can mean two things: the learner learned new words, or further reading revealed words they already knew, and our current model cannot tell the two apart. As a first step toward separating them, Figure 1 marks which known words the learner once looked up (the darker band): the only ones we saw them not know first. They are few, 1 in 25 known words, but a growing share of monthly gains: 3% in a learner’s first 20 articles, about 11% after 60. Even these are only candidates for learning, since a lookup is not always an actual gap (§6.2). For the words never looked up, it cannot say whether they were known all along or learned without help.

6. Validation

How far can a missing lookup be trusted? We test it from two angles: how reliably a run of clean encounters indicates that a word is known (§6.1), and how much weight a single lookup can bear (§6.2).

6.1 How Many Clean Encounters Make a Word Known?

We can quantify the rule directly. Of all words a learner met in N or more clean encounters **and then encountered again**, what share is never translated afterward? We call this share the *non-lookup rate*: the word, met again, never needed a translation afterwards. It measures behavioral consistency, not knowledge tested independently (§7). Conditioning on recurrence is needed: a word seen once and never met again would otherwise count as “known” for free, and roughly 70% of the single-encounter cases are exactly that. Across all learners (Table 4):

Clean encounters (N)	1	2	3	5	10	20
Non-lookup rate	92.0%	95.1%	95.8%	96.4%	97.1%	98.0%
Words	401,074	198,950	125,054	65,888	26,448	9,289

Table 4. Non-lookup rate after N or more clean encounters, among words encountered again afterwards.

A single clean encounter is followed by no later lookup in 92% of cases, and a second raises this to 95%; from there each doubling of clean encounters adds just under a point, reaching 98% at twenty. No single encounter settles the question, but each one counts, as the model assumes (§4.2), and the three-encounter threshold (§4.4) is therefore conservative: our model (§4.2) gives a word met cleanly three times a $P(\text{known})$ of only 0.79, well below the observed 96%. The rate stays below 100% because even well-known words are sometimes clicked (§6.2).

6.2 How Reliable Is a Single Lookup?

A **lookup** need not mean the learner did not know the word: it may be a passing confirmation of a word already known, or a misclick. And since Zeeguu also pronounces a word when it is clicked, a learner may click a word they know just to hear how it is said, especially in a language such as Danish, where spelling is a poor guide to pronunciation. In the extreme, one learner looked up the top-100 word *aber* (German “but”) only after reading it in 250 articles. And even a genuine lookup is only a snapshot: of the words a learner looked up and then met again, only **23% were looked up a second time**. For the rest, one lookup was enough, the word learned in the meantime, or never an actual gap to begin with.

7. Limitations

Our main limitations concern how far a missing lookup can be trusted as a sign of knowledge.

- **Reading is inferred, not observed.** Non-translation counts as knowledge only if the learner read the word, yet we cannot confirm they read the article or processed every word rather than skimming, so for an individual word the estimate is an *upper* bound (§9). Totals lean the other way (below).

- **Context can substitute for knowledge.** A word may go untranslated because the learner inferred it from context, not because they know it independently. What a missing lookup measures is that the learner did not need help to keep reading: functional receptive knowledge in context, not knowledge verified in isolation. A learner may also skip a lookup because they *believe* they know a word they do not, as with false friends; L2 readers routinely overestimate how well they understand unfamiliar words in a text (Laufer & Yano, 2001). Yes/No tests guard against such overconfidence with pseudowords (§3); our method has no equivalent check.
- **Multi-word expressions.** Multi-word lookups are set aside at the word level (§4), and expressions are not yet modeled as units, so the model cannot represent that a learner can know every component word yet not the combination (Danish *jo faktisk*, roughly ‘actually, you know’).
- **Polysemy.** Each stem has a single estimate, so its senses are merged: a learner who knows a word’s frequent sense is credited with knowing the word even where a text uses a rarer sense they do not know, and a lookup prompted by an unfamiliar sense counts against the word as a whole. Contextual embeddings could separate the senses (§9).
- **No external ground truth.** Underlying all of these, the non-lookup rates we report in §6 are measured against the learners’ own later behavior (whether they re-translate a word or leave it alone), not against an external test of what they know; only such a test would turn behavioral consistency into calibrated accuracy (§9).

The remaining limitations concern the coverage figures and their reach.

- **Growth may or may not be learning.** The estimate shows vocabulary demonstrated so far, not when it was learned: before the learner joined, or since. Only words once looked up and later read without help suggest learning during use (4% of known words), and even a lookup can be a mere check (§6.2).
- **Totals understate.** Vocabulary size and CEFR level lean low, for three reasons. Blocks of the frequency range in which a learner has met fewer than five stems are credited nothing (§4.4), so a light reader loses whole stretches of vocabulary they partly know. The *known* threshold is strict: it takes three clean encounters, although 92% of words are not looked up again after just one (Table 4). And the subtitle-based frequency lists (next bullet) rank conversational words high, but news articles rarely contain them, so they fall into those sparse blocks. Unread encounters (above) can inflate the estimate for words a learner has met, but cannot credit words never met.
- **Frequencies from the wrong register.** The frequency lists are built from film subtitles while learners read news, so conversational high-frequency words are under-encountered and coverage figures understate vocabulary (median Top-100 coverage across all profiles is 88%) rather than revealing a gap. A written-register list would raise them all.
- **Uncertainty is unquantified.** We report point estimates, and two sources of uncertainty go unshown. The extrapolation’s per-block sample sizes make sampling confidence intervals computable, which a longer version would report. And the absolute size depends on where the *known* threshold is set: counting a stem known at three clean encounters rather than two shifts every learner’s estimate by a sizable margin, because a word met exactly twice sits just under the cutoff. We report the conservative choice and read the CEFR level as indicative (§4.4).
- **A flat prior.** Every unseen word starts at $P(\text{known}) = 0.5$, using none of the indicators that could inform it (its frequency band, the learner’s self-declared level, or its cognateness with the first language); richer priors could fold these in at no cost in new data.
- **Generalizability.** The learners are self-selected, persistent, and French-dominated (§4.5), so the regularities we report may not extend to beginners, casual users, or the less-sampled languages.
- **Scope and granularity.** The method measures written reading vocabulary only (not listening, speech, or production) and counts stems, not lemmas (§4.4).

8. Applications

Even with all these limitations, we believe continuous vocabulary estimation can support several applications beyond placement testing.

Plugging vocabulary gaps. The estimate exposes a learner’s gaps: frequent words they have translated repeatedly, or that their reading has never put in front of them. Rather than wait for these to recur by chance, the platform can steer them into what the learner reads next. Zeeguu already simplifies articles on demand (§2); the same machinery could rank recommendations by how many gap words they contain and bias the simplification to keep or introduce them.

Prioritizing practice. Spaced-repetition schedulers, from the Leitner box (Leitner, 1972) to recent bandit-based systems (Meng, 2025), decide *when* to review a word but infer whether it is known only from their own exercises, so a word picked up through reading stays invisible to them. Reading-derived $P(\text{known})$ supplies that state directly, and practice can skip the words a learner demonstrably reads without help. In Zeeguu, a single lookup currently queues a word for practice regardless of the clean reading around it: it would queue a word like *aber* (§6.2), which $P(\text{known})$ keeps out.

A learner-facing dashboard. A dashboard of estimated vocabulary size, band coverage, and CEFR milestones over time (Figure 1) could provide motivational feedback, since visualizing progress is associated with increased engagement (Verbert et al., 2014). It should present the total as vocabulary demonstrated so far, and credit as learned only the words once looked up and now read without help (§7).

9. Conclusion and Future Work

Receptive vocabulary knowledge can be estimated from incidental reading behavior alone, without tests and without manual marking. Learners who need a translation ask for it early, most on the very first article containing the word and almost all by the second. So a word left untranslated after three or more clean encounters is not looked up again in about 96% of cases. Aggregated over a learner’s reading, the resulting per-word estimates yield a vocabulary estimate, updated monthly, for all 324 profiles in our sample.

Three directions would most strengthen the estimate. The first is validation against explicit vocabulary tests. The second is fitting the model’s weights and decay (w_{clean} , w_{lookup} , λ ; §4.2) to data as a time-based forgetting-curve or knowledge-tracing model, rather than hand-setting them. The third is measuring how much of an article was actually read, by combining the idle-aware timer (§4.1) with recently added scroll tracking; this would tighten the upper bound and likely lift the non-lookup rates, since a confirmed-read encounter is stronger evidence. Beyond these, contextual word embeddings could tell a word’s senses apart by clustering the contexts in which it occurs, so that knowledge is tracked per sense rather than per form.

10. Data and Code Availability

The anonymized dataset and the full extraction and analysis pipeline are openly available at <https://github.com/zeeguu/paper-vocabulary-estimation-eurocall-2026>. Learner identities are replaced by opaque codes (L01, L02, ...), with no names, emails, or free text.

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